Addition of angular momenta

$$\vec{J}_1 + \vec{J}_2 = \vec{J}$$

Naturally, the sum of angular momenta is also an angular momentum.

What if we consider $\vec{T} = a\,\vec{J}_1 + b\,\vec{J}_2$?

By definition, angular momentum $\vec{J}$ satisfies

$$[J_i, J_j] = i\hbar\, \varepsilon_{ijk} J_k$$

Let us check $\vec{T}$ now:

$$[T_i, T_j] = [a J_{1i} + b J_{2i}, a J_{1j} + b J_{2j}] =$$

$$= a^2 [J_{1i}, J_{1j}] + b^2 [J_{2i}, J_{2j}] = \left(\text{because } [\vec{J}_1, \vec{J}_2] = 0\right)$$

$$= a^2 i \varepsilon_{ijk} J_{1k} + b^2 i \varepsilon_{ijk} J_{2k} =$$

$$= i \varepsilon_{ijk} \left(a^2 J_{1k} + b^2 J_{2k}\right) \neq i \varepsilon_{ijk} \underbrace{\left(a J_{1k} + b J_{2k}\right)}_{T_k}$$

The only possibility : $a = b = 1$

(or $a = 1, b = 0$ or $a = 0, b = 1$)

<u>Eigenfunctions</u>

$$\left.\begin{aligned} \hat{J}_1^2 |j_1, m_1\rangle &= \hbar^2 j_1(j_1+1) |j_1, m_1\rangle \\ \hat{J}_1^z |j_1, m_1\rangle &= \hbar m_1 |j_1, m_1\rangle \end{aligned}\right\} \begin{aligned} &(2j_1+1) \text{ functions} \\ &-j_1 \le m_1 \le j_1 \end{aligned}$$

$$\left.\begin{aligned} \hat{J}_2^2 |j_2, m_2\rangle &= \hbar^2 j_2(j_2+1) |j_2, m_2\rangle \\ \hat{J}_2^z |j_2, m_2\rangle &= \hbar m_2 |j_2, m_2\rangle \end{aligned}\right\} \begin{aligned} &(2j_2+1) \text{ functions} \\ &-j_2 \le m_2 \le j_2 \end{aligned}$$

$\Rightarrow$ $\vec{J} = \vec{J}_1 + \vec{J}_2$ has a $D = (2j_1+1)\cdot(2j_2+1)$ dimensional space of functions $|j_1, m_1; j_2, m_2\rangle = |j_1, m_1\rangle |j_2, m_2\rangle$

Clearly $J^z = J_1^z + J_2^z$: $\hat{J}^z |j_1, m_1; j_2, m_2\rangle = \hbar(m_1 + m_2) |j_1, m_1; j_2, m_2\rangle$

thus eigenvalue of $\hat{J}^z$: $\hbar m$, with $m = m_1 + m_2$.

Note that $m_{max} = m_{1max} + m_{2max} = j_1 + j_2$

which defines unique state $|j_1, j_1; j_2, j_2\rangle$.

Since for arbitrary angular momentum $m_{max} = j$,

it follows that $|j_1, j_1; j_2, j_2\rangle$ is a state of angular momentum $j = j_1 + j_2$ $\Rightarrow$ hence $\hat{J}^2 |j_1, j_1; j_2, j_2\rangle = \hbar^2 (j(j+1)) \cdot |j_1, j_1; j_2, j_2\rangle$.

Then by acting on this state with $\hat{J}^- = \hat{J}_1^- + \hat{J}_2^-$ repeatedly, we generate $[2(j_1+j_2)+1]$ states, each of angular momentum $j = j_1 + j_2$ but with different m's:

$-(j_1+j_2) \le m \le j_1 + j_2$. Of all these states, only <u>one</u> has $m = j_1 + j_2 - 1$ (every $|m| \le j_1 + j_2$ occurs only once).

Such a state obtains as $\hat{J}^- |j_1, j_1; j_2, j_2\rangle =$

$$= (\hat{J}_1^- + \hat{J}_2^-) |j_1, j_1; j_2, j_2\rangle \sim \sqrt{2j_1}\, |j_1, j_1 - 1; j_2, j_2\rangle + \sqrt{2j_2}\, |j_1, j_1; j_2, j_2 - 1\rangle.$$

$$= |\varphi_+\rangle.$$

However there is another function,

$$|\varphi_-\rangle = \sqrt{2j_2}\, |j_1, j_1 - 1; j_2, j_2\rangle - \sqrt{2j_1}\, |j_1, j_1; j_2, j_2 - 1\rangle$$

Note that $\langle \varphi_+ | \varphi_- \rangle = 0$. Note also that

$$\hat{J}^z |\varphi_-\rangle = (\hat{J}_1^z + \hat{J}_2^z) |\varphi_-\rangle = \hbar (j_1 - 1 + j_2) \sqrt{2j_2}\, |j_1, j_1 - 1; j_2, j_2\rangle$$

$$- \hbar (j_1 + j_2 - 1) \sqrt{2j_1}\, |j_1, j_1; j_2, j_2 - 1\rangle = \hbar (j_1 + j_2 - 1) |\varphi_-\rangle.$$

Hence $m_{\varphi_-} = j_1 + j_2 - 1$.

Also, $\hat{J}^+ |\varphi_-\rangle = (\hat{J}_1^+ + \hat{J}_2^+) |\varphi_-\rangle \sim \sqrt{2j_2 \cdot 2j_1}\, |j_1, j_1; j_2, j_2\rangle - \sqrt{2j_1 \cdot 2j_2}\, |j_1, j_1; j_2, j_2\rangle$

$$= 0.$$

$\Rightarrow$ $\max(m_{\varphi_-}) = j_1 + j_2 - 1$ $\Rightarrow$ this function belongs to $j = j_1 + j_2 - 1$ manifold.

$J^- |\ell, m\rangle = \hbar \sqrt{(\ell - m + 1)(\ell + m)}\, |\ell, m - 1\rangle$

$J^+ |\ell, m\rangle = \hbar \sqrt{(\ell - m)(\ell + m + 1)}\, |\ell, m + 1\rangle$

Hence by applying $\hat{J}^-$ to $|\varphi_-\rangle$ repeatedly, we will generate $[2(j_1+j_2-1)+1]$ different values of m.

Look now on $m = j_1+j_2-2$ which can be achieved in 3 ways: $m_1 = j_1-2,\ m_2 = j_2$; $m_1 = j_1-1,\ m_2 = j_2-1$; $m_1 = j_1,\ m_2 = j_2-2$. Two possibilities have already being accounted for via $|\varphi_+\rangle$ and $|\varphi_-\rangle$ states. The <u>third</u>, orthogonal to both $|\varphi_\pm\rangle$, can be used to construct the manifold with $j = j_1+j_2-2$.

This procedure can be continued until all $(2j_1+1)(2j_2+1)$ original states have been used. This happens when $j = |j_1-j_2|$. (Indeed, assume $j_1 > j_2$ for definiteness: than the minimal possible value of $m_{max} = j_1-j_2$. Hence the "minimal" manifold must have $j = j_1-j_2$.)

Thus, upon adding $\vec{J}_1$ (with $\hat{J}_1^2 \to \hbar^2 j_1(j_1+1)$) and $\vec{J}_2$ (with $\hat{J}_2^2 \to \hbar^2 j_2(j_2+1)$) we get ~~states~~ manifolds with angular momenta $\boxed{j_1-j_2 \le j \le j_1+j_2}$. Each has $(2j+1)$ ~~degenerate~~ with different m's.

Total number of them $\sum_{j_1-j_2}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1)$.

Check it: $\sum_{j_1-j_2}^{j_1+j_2} (2j+1) = \underbrace{2(j_1-j_2)+1}_{1} + \underbrace{2(j_1-j_2+1)+1}_{2} + \ldots + \underbrace{2(j_1+j_2)+1}_{2j_2+1}$

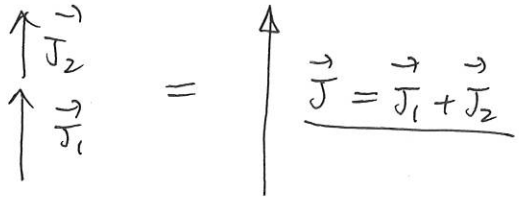
(the sum has $(2j_2+1)$ terms) $=$

$$= (2j_1)\cdot(2j_2+1) + (1)\cdot(2j_2+1) = (2j_1+1)(2j_2+1).$$

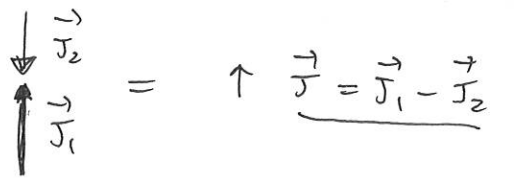
[Note that every entry with $-j_2+k$ $(0 \le k \le 2j_2+1)$ is compensated by a symmetric one with j_2-k.)

Very clear classical analogy: when adding vectors $\vec{J}_1$ and $\vec{J}_2$, the maximum length vector has $j = j_1 + j_2$,
(j_1) (j_2) the minimum length $j = j_1 - j_2$.

<u>parallel</u>



<u>antiparallel</u>



Denote the new set of states as
$|j, m; j_1, j_2\rangle$.

$$|j_1 - j_2| \leq j \leq j_1 + j_2$$

Note that $\hat{J}^2 |j, m, j_1, j_2\rangle = \hbar^2 j(j+1) |j, m, j_1, j_2\rangle$

$$\hat{J}^z |j, m, j_1, j_2\rangle = \hbar m |j, m, j_1, j_2\rangle$$

$$-j \leq m \leq j$$

Note that $[\vec{J}_1^{\,2}, \vec{J}^{\,2}] = [\vec{J}_1^{\,2}, (\vec{J}_1 + \vec{J}_2)^2] = 0$

(because for any $\vec{L}$: $[\vec{L}^2, L^{x,y,z}] = 0$
angular momentum squared commutes with every component of $\vec{L}$.)

$\Rightarrow$ these states are also eigenstates of $\hat{J}_1^2$ and $\hat{J}_2^2$: $\hat{J}_1^2 |j, m, j_1, j_2\rangle$
$= \hbar^2 j_1(j_1+1) |j, m, j_1, j_2\rangle$

Quite generally:

$$|j, m, j_1, j_2\rangle = \sum_{m = m_1 + m_2} C^{j_1 j_2 j}_{m_1 m_2 m} \, |j_1, m_1; j_2, m_2\rangle$$

$$\Downarrow$$

$\langle j_1, m_1; j_2, m_2 | j, m, j_1, j_2\rangle$ = Clebsch–Gordan coefficients

Explicit construction for $J_1 = J_2 = \frac{1}{2}$; $J_1 = 1$, $J_2 = \frac{1}{2}$; $J_1 = l$, $J_2 = \frac{1}{2}$.

① $J_1 = \frac{1}{2}$, $J_2 = \frac{1}{2}$ $\Rightarrow$ $0 = \frac{1}{2} - \frac{1}{2} \leq J \leq \frac{1}{2} + \frac{1}{2} = 1$

$\Rightarrow$ 4 states: 3 with $J = 1$ ($m = -1, 0, 1$) and 1 with $J = 0$ ($m = 0$).

<u>Original</u> states are $|\frac{1}{2}, \pm\frac{1}{2}; \frac{1}{2}, \pm\frac{1}{2}\rangle \xrightarrow[\text{simplify notation}]{} |\psi_{\pm\frac{1}{2}, \pm\frac{1}{2}}\rangle$

<u>New states</u> are eigenstates of total $\vec{J}^2$ and J_z (as well as $\vec{J}_1^2, \vec{J}_2^2$)

Denote as $|\phi_{j,m}\rangle$ for brevity.

Clearly $|\phi_{1,1}\rangle = |\psi_{\frac{1}{2}, \frac{1}{2}}\rangle$ as there is only 1 state with $m = 1 = \frac{1}{2} + \frac{1}{2} = m_1 + m_2$.

To find $|\phi_{1,0}\rangle$ we act with $\hat{J}_-$ on $|\phi_{1,1}\rangle$:

By def $\hat{J}_- |\phi_{1,1}\rangle = \sqrt{j(j+1) - m(m-1)}\, |\phi_{1, \underbrace{1-1}_{0}}\rangle = \sqrt{2}\, |\phi_{1,0}\rangle$

But in the old basis

$$\hat{J}_- |\phi_{1,1}\rangle = (\hat{J}_{1-} + \hat{J}_{2-}) |\psi_{\frac{1}{2}, \frac{1}{2}}\rangle = |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle + |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle$$

$$\Rightarrow |\phi_{1,0}\rangle = \frac{1}{\sqrt{2}} \left(|\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle + |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle \right).$$

Then $\hat{J}_- |\phi_{1,0}\rangle = \sqrt{2 - 0}\, |\phi_{1,-1}\rangle = \sqrt{2}\, |\phi_{1,-1}\rangle$

but $\hat{J}_- |\phi_{1,0}\rangle = (\hat{J}_{1-} + \hat{J}_{2-}) \frac{1}{\sqrt{2}} \left(|\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle + |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle \right) =$

$$= \frac{1}{\sqrt{2}} 2 |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle = \sqrt{2}\, |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle$$

$$\Rightarrow |\phi_{1,-1}\rangle = |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle.$$

The remaining state $|\phi_{0,0}\rangle$ must be orthogonal to $|\phi_{1,0}\rangle$

$$\Rightarrow |\phi_{0,0}\rangle = \frac{1}{\sqrt{2}} \left(|\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle - |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle \right).$$ <u>Singlet</u>

by inspection

Naturally $|\phi_{1,m}\rangle$ = triplet states.

Spin-orbit interaction in atoms:

$H_{SO} = \lambda \vec{S}\cdot\vec{L} \Rightarrow \vec{L}$ and $\vec{S}$ are not separately conserved, but $\vec{J} = \vec{L}+\vec{S}$ is.

$$H_Z = -\vec{\mu}\cdot\vec{B} \quad , \vec{\mu} = \frac{\mu_B}{\hbar}\left(\vec{L} + 2\vec{S}\right) \quad (1)$$

$$\vec{\mu} = \frac{\mu_B}{\hbar} g_J \vec{J} \quad (2)$$

So need to find projection of $\vec{L}+2\vec{S}$ onto $\vec{J} = \vec{L}+\vec{S}$, it should be conserved.

$$\vec{\mu}\cdot\vec{J} = \frac{\mu_B}{\hbar}\left(\vec{L}\cdot\vec{J} + 2\vec{S}\cdot\vec{J}\right)$$

$$\vec{\mu}\cdot\vec{J} = \frac{\mu_B}{\hbar} g_J \vec{J}\cdot\vec{J} = \frac{\mu_B}{\hbar} g_J \hbar^2 J(J+1)$$

$$\vec{L} = \vec{J}-\vec{S} \Rightarrow \vec{L}^2 = (\vec{J}-\vec{S})^2 = \vec{J}^2 + \vec{S}^2 - 2\vec{S}\cdot\vec{J}$$

$$\Rightarrow \vec{S}\cdot\vec{J} = \frac{1}{2}\left(\vec{J}^2 + \vec{S}^2 - \vec{L}^2\right)$$

$$\vec{S} = \vec{J}-\vec{L} \Rightarrow \vec{S}^2 = \vec{J}^2 + \vec{L}^2 - 2\vec{J}\cdot\vec{L}$$

$$\Rightarrow \vec{J}\cdot\vec{L} = \frac{1}{2}\left(\vec{J}^2 + \vec{L}^2 - \vec{S}^2\right)$$

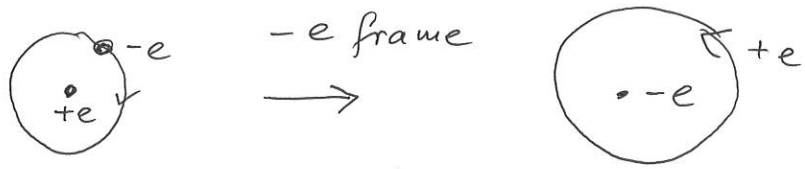
Thus

$$\frac{\mu_B}{\hbar}\left\{\frac{1}{2}\hbar^2\left(J(J+1) + L(L+1) - S(S+1)\right) + 2\cdot\frac{1}{2}\hbar^2\left(J(J+1) + S(S+1) - L(L+1)\right)\right\} = \frac{\mu_B}{\hbar} g_J \hbar^2 J(J+1)$$

$$\Rightarrow g_J J(J+1) = \frac{3}{2}J(J+1) - \frac{1}{2}L(L+1) + \frac{1}{2}S(S+1)$$

$$g_J = \frac{3}{2} - \frac{L(L+1) - S(S+1)}{J(J+1)}$$

Spin-orbit coupling



proton $(+e)$ produces current $I = e/\tau$

$\Rightarrow$ magnetic field $\vec{B} = \dfrac{\mu_0 \vec{I}}{2r}$

Electron's orbital momentum $L = rmv = rm\dfrac{2\pi r}{\tau} = 2\pi \dfrac{mr^2}{\tau}$

$$\Rightarrow B = \frac{\mu_0}{2r}\frac{e}{\tau} = \frac{\mu_0 e}{4\pi m r^3} L \qquad \frac{1}{\tau} = \frac{L}{2\pi m r^2}$$

$$\sqrt{\mu_0 \varepsilon_0} = \frac{1}{c} \Rightarrow \mu_0 = \frac{1}{c^2 \varepsilon_0}$$

$$\vec{B} = \frac{1}{4\pi\varepsilon_0}\frac{e}{mc^2 r^3}\vec{L}$$

$\Rightarrow$ electron's ~~spin~~ magnetic moment interacts with that $\vec{B}$:

$$H_{so} = -\vec{\mu}_e \cdot \vec{B} \quad , \vec{\mu}_e = \frac{g(-e)}{2m}\vec{S} = -\frac{e}{m}\vec{S} \quad (g=2)$$

$$\Rightarrow H_{so} = \frac{e^2}{4\pi\varepsilon_0}\frac{1}{m^2c^2r^3}\vec{S}\cdot\vec{L} \qquad \text{spin-orbit coupling}$$

Caveat: the frame we used is not <u>inertial</u> $\Rightarrow$ Thomas correction gives factor $(1/2)$ — follows from Dirac's eq.

$$H_{so} = \frac{e^2}{8\pi\varepsilon_0}\frac{1}{m^2c^2r^3}\vec{S}\cdot\vec{L} = \lambda\,\vec{S}\cdot\vec{L}$$

For hydrogen-like atom of charge Ze : $\lambda = \dfrac{Ze^2}{8\pi\varepsilon_0 m^2c^2r^3}$

Moreover $\left\langle \dfrac{1}{r^3}\right\rangle = \dfrac{Z^3}{a_0^3 n^3 \, l(l+\frac{1}{2})(l+1)}$

$\Rightarrow \lambda \sim Z^4$: SO becomes strong in heavy atoms.

Another point of view (very closely related):

orbiting nucleus $(+e)$ creates magnetic field

$$\vec{B} = \frac{\vec{E} \times \vec{v}}{c^2}, \quad \vec{E} = -\vec{\nabla} V(r) = -\hat{r}\,\frac{dV(r)}{dr}$$ is the electric field of $(+e)$

[Griffiths E&M, Ch. 12]

$$= -\frac{\vec{r}}{r}\,\frac{dV(r)}{dr}$$

$$\vec{B} = -\frac{1}{r}\frac{dV}{dr}\,\frac{\vec{r}\times\vec{v}}{c^2} = \frac{1}{r}\frac{dV}{dr}\,\frac{m\vec{v}\times\vec{r}}{mc^2} = \frac{1}{r}\frac{dV}{dr}\,\frac{\vec{L}}{mc^2}$$

But $V(r) = -\dfrac{Ze^2}{4\pi\varepsilon_0}\dfrac{1}{r}$

$$\vec{B} = \frac{Ze^2}{4\pi\varepsilon_0 r^3}\,\frac{\vec{L}}{mc^2}$$

$$\Rightarrow \quad H_{so} = -\underbrace{\left(\tfrac{1}{2}\right)}_{\text{Thomas factor}} \vec{\mu}_e \vec{B} = \frac{Ze^2}{8\pi\varepsilon_0 m^2 c^2 r^3}\,\vec{S}\cdot\vec{L}$$ as before.

Thus $H_{so} = \lambda \vec{S}\cdot\vec{L}$

expectation value $\langle H_{so}\rangle = \lambda\langle \vec{S}\cdot\vec{L}\rangle = \cancel{\frac{\lambda}{2}\langle \vec{J}^2 + \vec{S}^2 - \vec{L}^2\rangle =}$

$\cancel{= \frac{\lambda}{2}\left[J(J+1) + S(S+1) - L(L+1)\right]}$

$$\vec{J} = \vec{S} + \vec{L}$$

$$\Rightarrow \vec{J}\cdot\vec{J} = \vec{S}^2 + \vec{L}^2 + 2\vec{S}\cdot\vec{L}$$

$$\vec{S}\cdot\vec{L} = \frac{1}{2}\left(\vec{J}^2 - \vec{L}^2 - \vec{S}^2\right)$$

$$\Rightarrow \langle H_{so}\rangle = \frac{\lambda}{2}\left(J(J+1) - L(L+1) - S(S+1)\right)$$